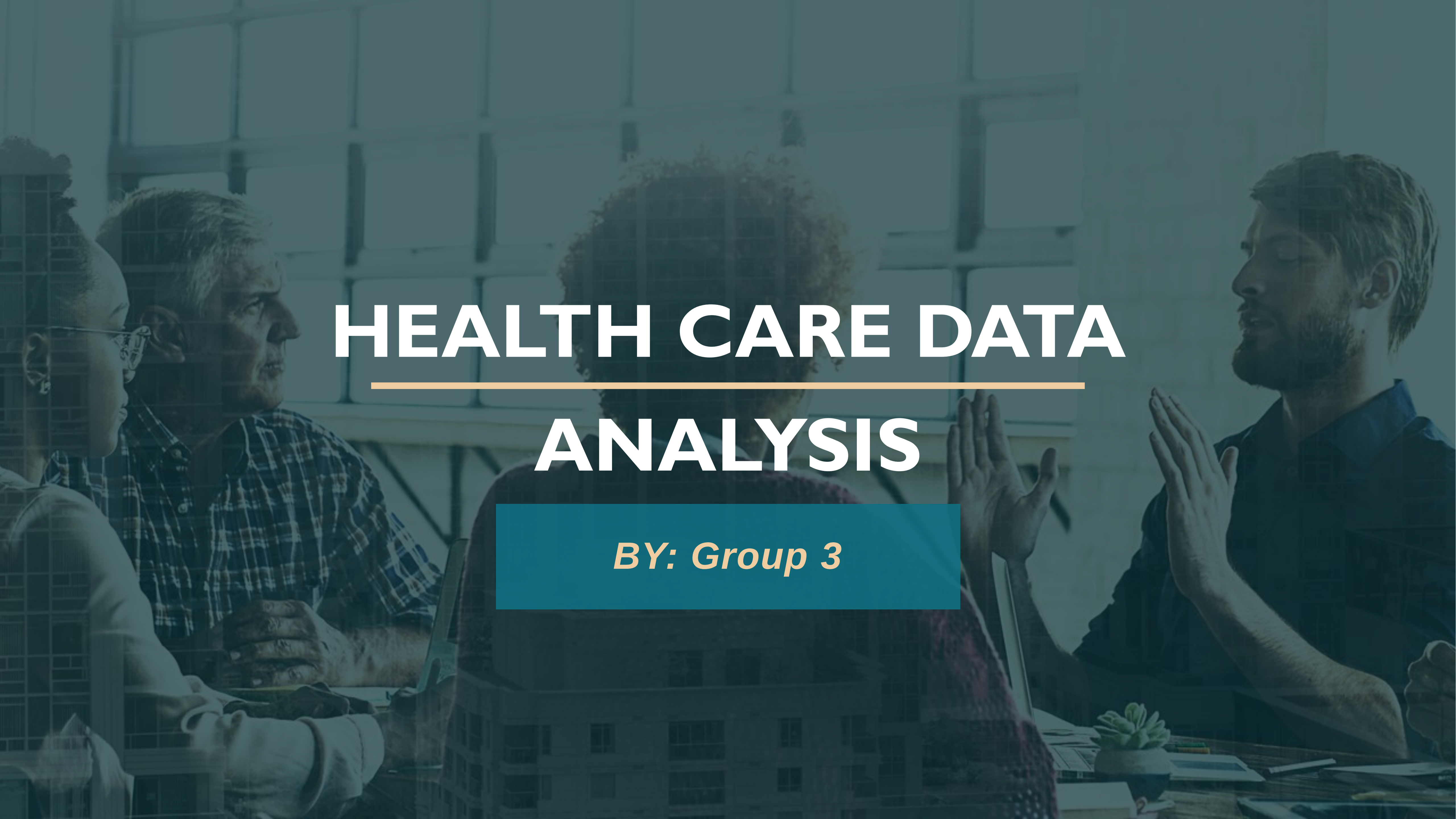


HEALTH CARE DATA ANALYSIS

BY: Group 3

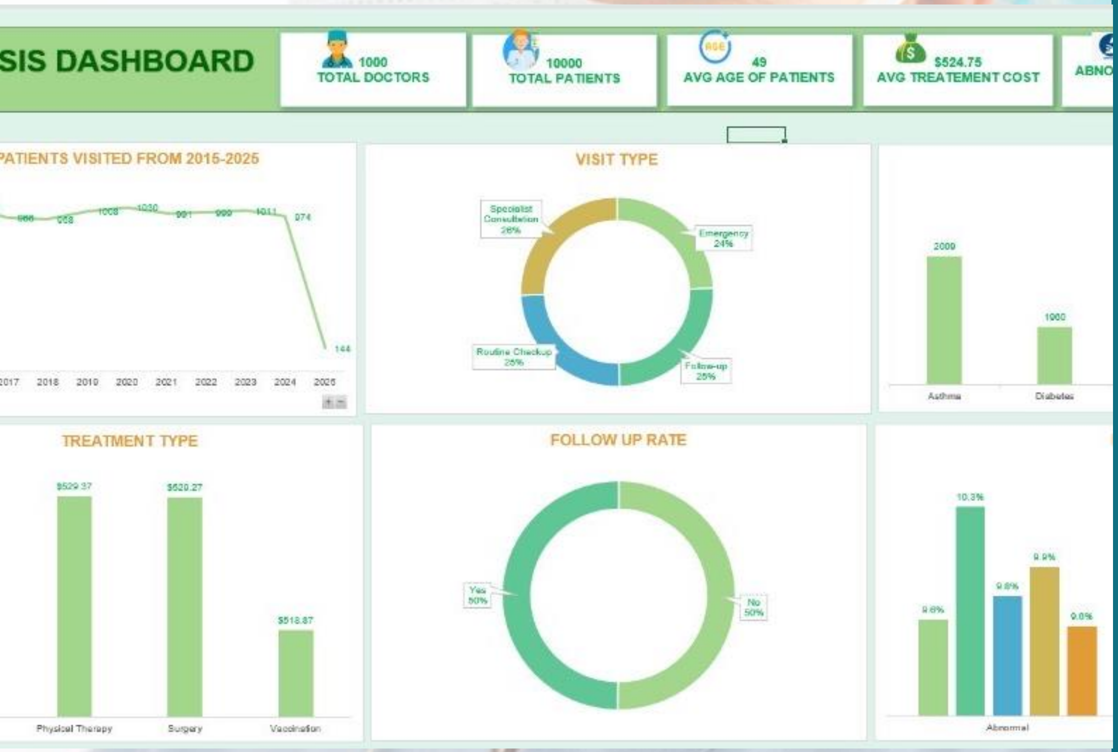
PROJECT SUMMARY

To analyze healthcare data across various metrics including patient visits, diagnoses, treatments, test results, and doctor workload and to use the master tools like Excel , Power BI , Tableau and SQL. With the scope of comparative analysis across cities, test types, visit types, and insurance providers,

KEY PERFPRMANCE INDICATORS

- AVG Patient Age: 49
- Total Doctors: 1,000
- Total Patient: 10,000
- Total Visit: 10,000
- Doctor Workload: 10 patient
- Most Common Visit Type: Specialist Consultation (25.59%)
- Top Diagnosis: Migraine (20.39%)
- Year with most visits: 2024 (4,927 visits)
- AVG Treatment cost: $518.87-$529.27

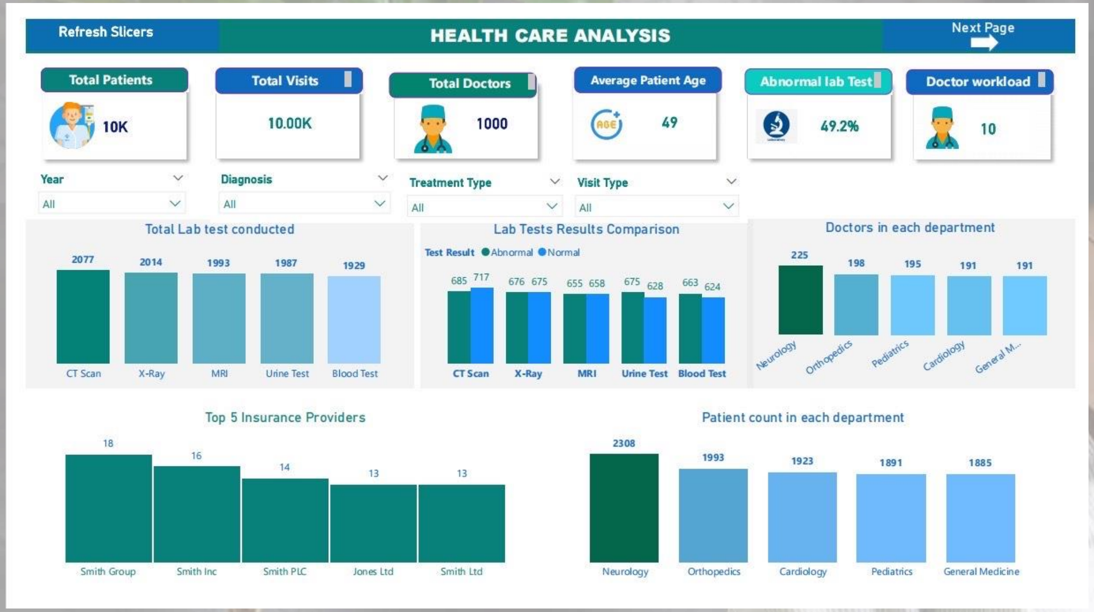
Excel Dashboard

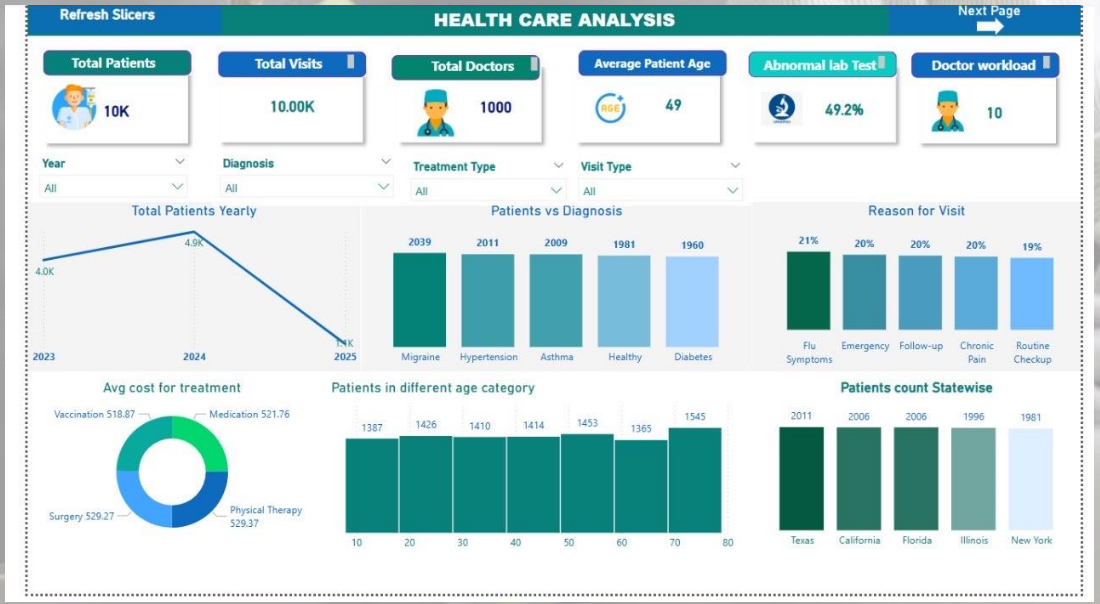


Insights from Dashboard

- Distribution of visit types and diagnoses
- High follow-up rate and abnormal test outcomes
- Lab results indicate potential quality control focus

Tableau Dashboard

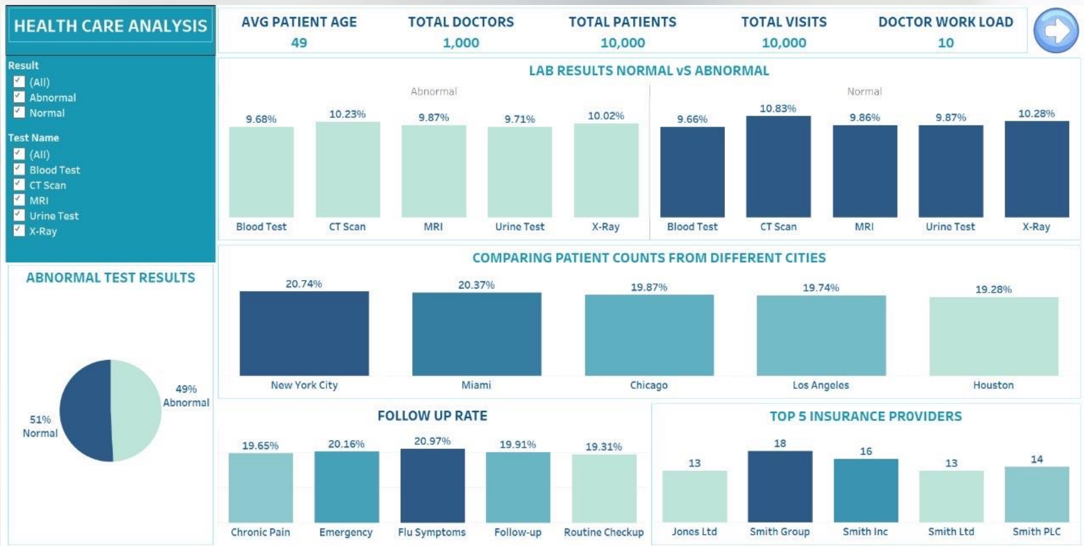


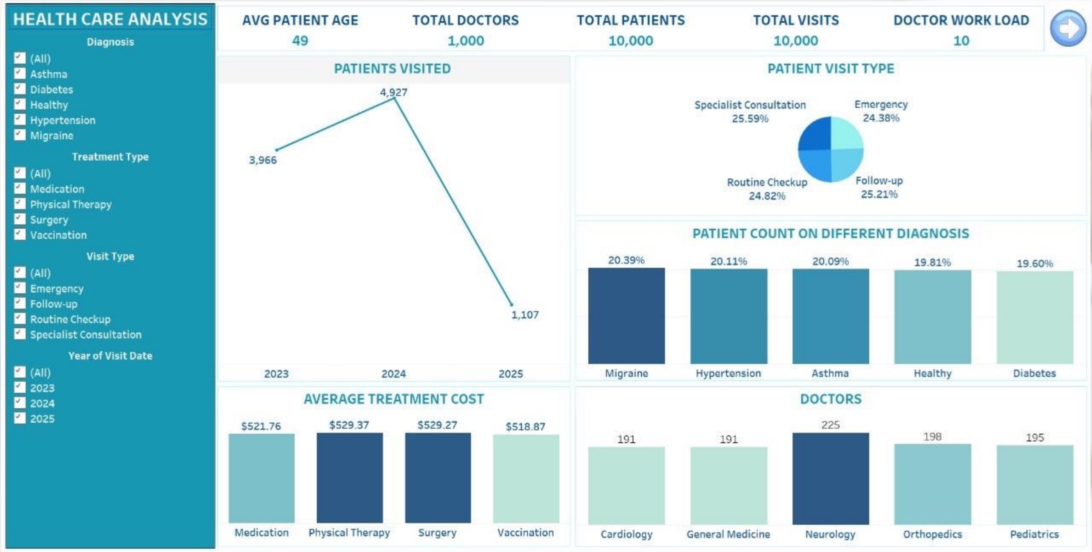


Insights from Dashboard

- Abnormal lab test rate: 49%
- Department-wise doctor distribution
- Cost comparison of top insurance providers
- Regional patient distribution analysis

Power BI Dashboard

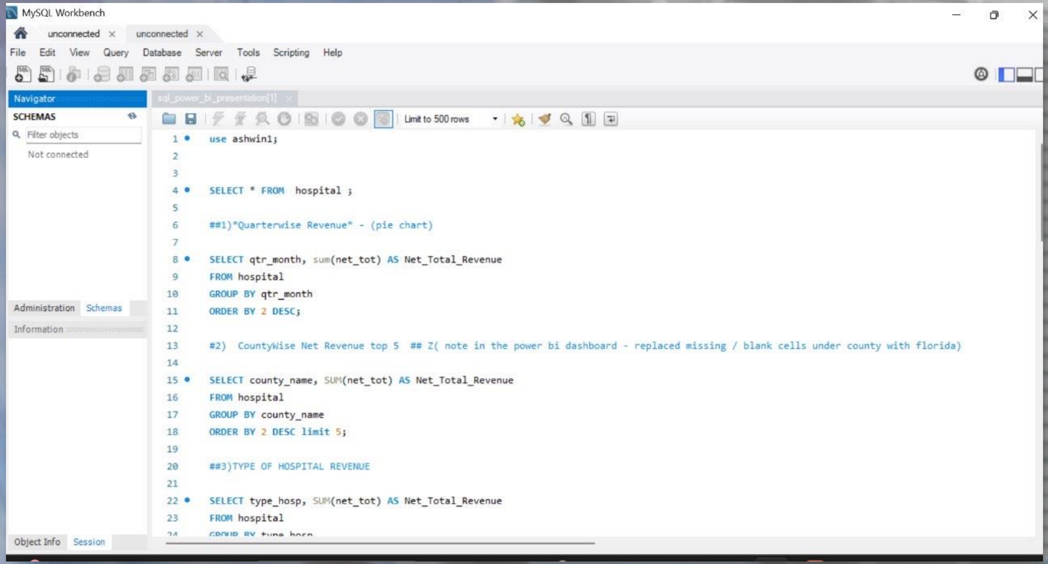


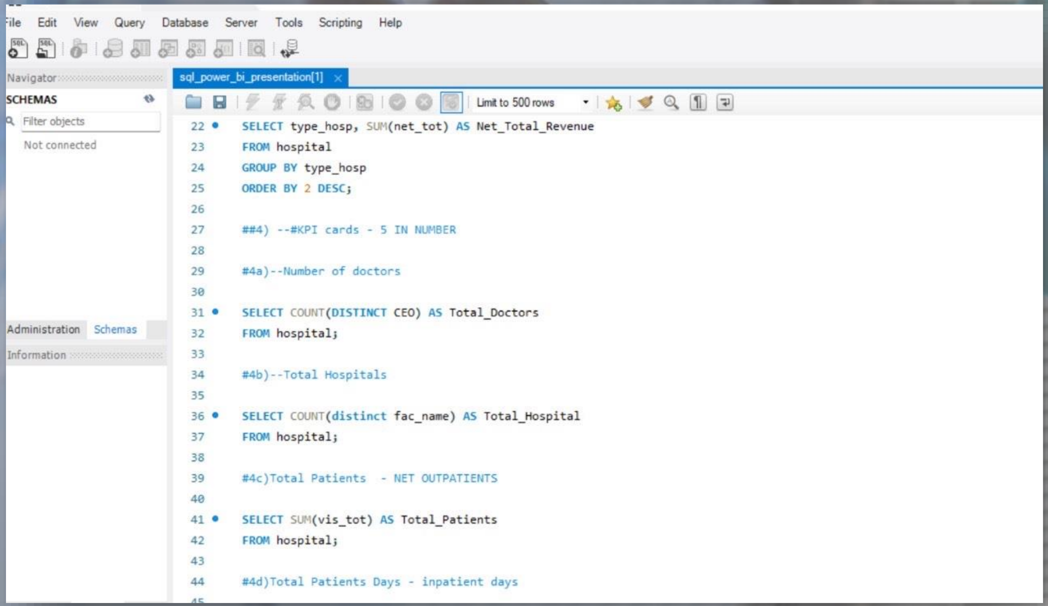


Insights from Dashboard

- Top cities with high patient volume: NYC &Miami
- Treatment cost variation across diagnosis types
- High patient count for chronic illnesses
- Deep dive into follow-up frequency

SQL Query

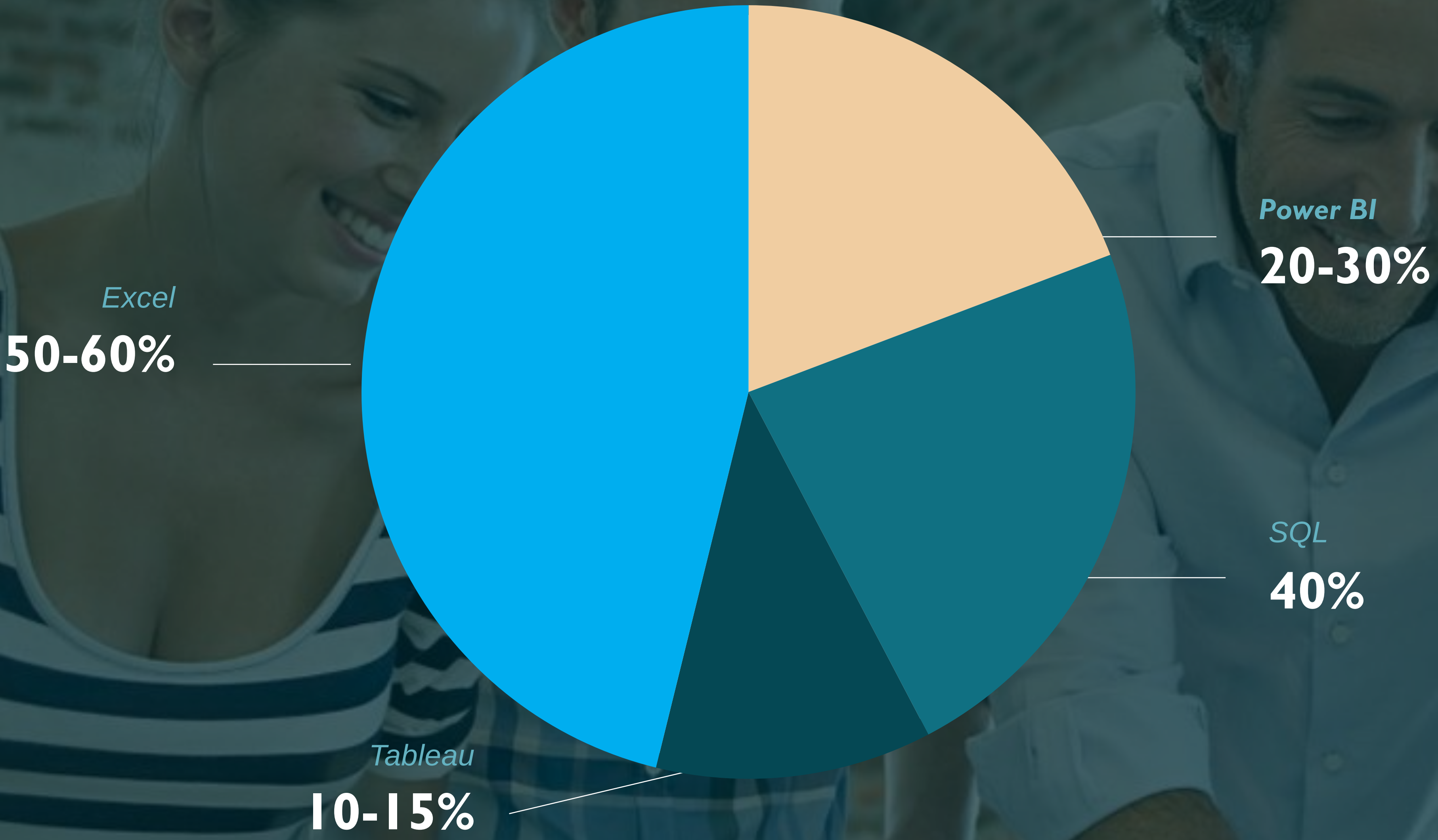




Insights from Dashboard

- Patient age & workload
- Top diagnoses & treatment cost
- Follow-up rates & abnormal test ratios
- Lab test distribution

USE OF TOOLS



Key Insights & Recommendations

- **Demographics:**
  Average patient age is 49, indicating a balanced population with both preventive and chronic care needs.

- **Workload Efficiency:**
  ~10 patients per doctor suggests manageable caseloads, supporting timely care and provider sustainability.

- **Follow-Up Visits (25%):**
  A solid portion of visits are follow-ups, reflecting good continuity of care but also ongoing patient management need

- **Abnormal Test Results (49%):**
  High rate of abnormal results may point to elevated health risks in the population, requiring closer clinical oversight.

- **Top Diagnoses:**
  Migraine, hypertension, and asthma are the most common—highlighting a need for focused chronic care pathways.

- **Recommendation:**
  Prioritize resource allocation in NYC and Miami due to higher demand or clinical burden.

THE TEAM

- Gayathry Prasad
- Nikhat Jarin
- Akul Srujanasheela

Group 3

- Komal Doolani
- Surabhi Basant Sahoo

THANK YOU!